Rugved Zarkar

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EDUCATION

University of Maryland, College Park Expected May 2028
Bachelor of Science in Computer Science, GPA: 3.83/4.00 *College Park, MD*
- **Relevant Coursework:** Operating Systems, Computer Networks, Database Design, Algorithms, Advanced Data Structures

TECHNICAL SKILLS

Languages: Python, Go, TypeScript, Java, JavaScript, SQL, C, C++, HTML/CSS
Infrastructure & Cloud: AWS (EC2, ECS, Lambda, DynamoDB, CloudWatch, S3, IAM, VPC, Route 53), Docker, Kubernetes, Terraform, Linux, Git, Bash, Ansible, Grafana
Backend & Frameworks: REST APIs, PostgreSQL, MySQL, Aurora, FastAPI, Node.js, Temporal, OpenTelemetry, Weaviate, LlamaIndex, React, PyTorch, TensorFlow, Scikit-Learn, LangChain
Certifications: AWS Certified Cloud Practitioner, AWS Certified Solutions Architect

PROFESSIONAL EXPERIENCE

Amazon Web Services Sep 2026 – Present
Software Engineer Intern *Arlington, VA*
- Developing a Go domain-classification API for AI tool calls, integrating Route 53 DNS Firewall and Bedrock AgentCore Policy with per-feature confidence scores, cache TTLs, and TypeScript/CDK infrastructure.
- Consolidated API and PrivateLink designs across 4 teams; identified authentication conflicts and retry paths that could exceed a 50 ms policy timeout before implementation.
- Built Route 53 DNS Firewall and Profiles analytics in TypeScript, Python, and Java across 36 AWS commercial regions, replacing 2–3 days of manual review with daily usage, growth, churn, cohort, and payer-level metrics.
- Built ECS Fargate ingestion from 72 cross-account snapshots with IAM/STS and DynamoDB; added AWS CDK pipelines, integration tests, and CloudWatch alarms with data isolation across commercial, GovCloud, and China partitions.

IBM May 2026 – Aug 2026
Software Engineer Intern *Austin, TX*
- Architected DNSDist/Unbound/SDNLB infrastructure for IBM's Classic-to-VPC migration, scaling throughput 10x from 100K to 1M+ QPS with sub-1% packet loss across 30 clients, 2 DNSDist nodes, and 3 Unbound nodes.
- Raised DNSDist throughput 19% through CPU pinning, SO_REUSEPORT, kernel scheduling, and network queue tuning.
- Built Python exporters and Grafana/Osprey dashboards for per-node DNS latency, health, packet loss, timeouts, and throughput.
- Cut DNS node setup from days to under 3 minutes with Ansible/systemd provisioning, exporters, and health checks.

University of Maryland Institute for Advanced Computer Studies Aug 2024 – Present
Software Research Assistant *College Park, MD*
- Developing C++/Python firmware for Calico, an ESP32-S3 wearable robot, connecting touchscreen controls, a 20+ Hz IMU, Wi-Fi, and real-time visualization while keeping touch-to-device command latency below 100 ms.
- Extended ACM IMWUT-published Calico for classrooms and CHI/UbiComp demos; featured in IEEE Spectrum and The Verge.

RAGent LLC Jan 2025 – Apr 2026
Founder & Lead Software Developer *Ellicott City, MD*
- Built an AWS RAG platform for 350 registered users using Terraform, Aurora v2, Bedrock Titan, and Claude 3.5 Haiku; processed 7,000+ restricted documents, cutting review time 75%, with government pilots in Ghana and Kenya.
- Built agentic RAG over a 150-node education framework and 1,000+ research sources to prioritize system improvements.
- Led an 11-person team; platform assets acquired by sustainED; technology presented at the UN in Sept 2026.

National Institute of Standards and Technology May 2025 – Mar 2026
Software Engineer Intern *Gaithersburg, MD*
- Built PyTorch metrology pipelines for 1,000+ SEM images, cutting scan points 40% and measurement uncertainty 50%.
- Fine-tuned diffusion models for SEM restoration and evaluated reconstruction quality against baseline using PSNR and SSIM.

PROJECTS

Claude Command Center | *TypeScript, Obsidian, Node.js, SQLite, macOS* Sept 2026
- Built an Obsidian plugin and Node.js service unifying Claude Code sessions, tasks, and research; implemented knowledge graphs to retain context and connect project knowledge across sessions for long-running projects.
- Built typed APIs with shared Zod schemas and SQLite; added collectors, schedulers, launchers, and macOS tool integrations.
- Secured Unix-domain sockets with HMAC authentication and Keychain secrets; added launchd supervision, atomic Markdown vault writes, versioned database migrations, and event replay/reconnection for reliable local services.
- Automated end-to-end tests and CI checks to detect exposed secrets, privacy issues, and package-boundary violations.

HarnessFlow: AI Agent Harness | *Go, Python, PostgreSQL, Temporal, Next.js* May 2026
- Built a Go/PostgreSQL service converting PRDs and Git repos into implementation branches with persisted plans and evidence.
- Orchestrated Python/Codex workers with Temporal, isolated Git worktrees, bounded repair loops, and pause/resume controls.
- Gated completion on frozen acceptance checks at the delivered commit, independent review, and per-requirement LLM approval.
- Built a Next.js dashboard and CLI to inspect branches, diffs, reports, and run history during implementation reviews.